Aditya Meenakshisundaram

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Education

Texas A&M University Expected May 2028
Bachelor of Science in Chemical Engineering
Cumulative GPA: 3.91 / 4.00 | Craig & Galen Brown Engineering Honors

Work Experience & Research

Photolithography & Metrology Intern May 2026 – August 2026
Samsung Semiconductor Austin, TX

- Recovered 300–400 wafers/day of throughput using I-MR statistical process control to isolate a downstream track bottleneck caused by a degraded buffer-unit transfer mechanism
- Built a Python-based overlay-correction tool that transformed 500+ wafer- and shot-level metrology measurements per lot into scanner alignment corrections, saving 250 engineering hours annually
- Developed and deployed a PyTorch CNN on FFT-derived heatmaps to classify spin-coater defects with 84% accuracy, automating review of 1,000+ previously manual process images per day
- Automated lot-level overlay metrology skip-factor resets with Python and SQL, restoring process-specific defaults while saving 50 engineering hours per week

Samsung Semiconductor Research Fellow Jan 2026 – Present
Samsung Semiconductor & Artie McFerrin Department of Chemical Engineering College Station, TX

- Developed a passive thermal-switching composite for lithium-ion battery thermal-runaway mitigation, using indium phase change, $CaCO_3$ decomposition, and a retained titanium framework to transition from heat conduction to insulation
- Calculated thermal conductivity ($k = \alpha \rho C_p$) across 50–70 wt% indium formulations, identifying 65 wt% as the strongest-performing composition across thermal response and structural integrity
- Fabricated composites end to end through powder mixing, pellet pressing, and 900 °C heat treatment; analyzed cross-sectional morphology and conductive pathways by SEM across 4+ prototype iterations
- Built a verified 1D finite-volume model of a 5-cell battery stack in Python; swept 361 barrier designs and identified a simulation-predicted hot-state conductance threshold of $\sim 32\,\mathrm{W\,m^{-2}K^{-1}}$ for suppressing runaway propagation

Undergraduate Researcher, Green Lab Aug 2025 – Present
Artie McFerrin Department of Chemical Engineering, Texas A&M University College Station, TX

- Project: conversion of liquid polycarbosilane (PCS) into electrically conductive silicon carbide for RF susceptor applications; targeted the lowest pyrolysis temperature needed to obtain conductivity
- Processed liquid PCS through degassing, curing, resting, and pyrolysis; completed 10+ controlled runs at 800–1200 °C while varying temperature, ramp rate, and atmosphere
- Narrowed the onset of measurable electrical conductivity to 900–1000 °C using four-point probe measurements, guiding subsequent trials within the identified transition range
- Achieved 92% SiC phase purity using XRD, examined backbone-fiber and PCS-coating cross-sections using SEM, and evaluated RF heating at 200–300 W to assess electromagnetic coupling and temperature response

Projects

STEM Research Finder Co-Creator; Lead Developer
stemresearchfinder.tech

- Co-created and launched a multi-tenant research-discovery platform supporting university-specific deployments; indexed 5,000+ STEM faculty nationwide and attracted 3,000+ student users without paid acquisition
- Built Python data pipelines across 50+ university web sources to extract and standardize faculty research profiles, deduplicate records, and automate index updates

Substack Author Semiconductor & Technology Writer
adimeenak.substack.com

- Published 30+ articles on semiconductors, materials, energy, and emerging technologies, reaching 400+ monthly readers

Honors & Technical Skills

Honors & Awards: Humba Deep Tech Fellowship, Samsung Semiconductor Research Fellowship, Samsung Semiconductor Intern Scholarship, TEX-E Fellowship, Omega Chi Epsilon
Characterization: Scanning Electron Microscopy, X-Ray Diffraction, Thermo-gravimetric Analysis, Optical Microscopy
Engineering Modeling: COMSOL Multiphysics, Aspen Plus, LOPA, Process Hazard Analysis, Fusion 360
Programming & Data: Python (NumPy, Pandas, SciPy, PyTorch, TensorFlow), SQL, MATLAB, Git, Docker, Linux